effect felt more freely. Above a certain amplitude of vibration, it is able to increase still further the vibration initiated, and eventually to tear the nucleus apart.



Figure 26.—Nuclear fission.

This fission of the nucleus may take place in various ways. As a rule, the two fragments are not of equal size. For instance, when the rare uranium isotope $_{92}U^{235}$ has absorbed a neutron, it may split into a strontium atom, $_{38}Sr^{90}$, and a xenon atom, $_{54}Xe^{144}$, and two neutrons, i.e.:

$$_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{38}Sr^{90} + {}_{54}Xe^{144} + {}_{0}n^{1} + {}_{0}n^{1}$$

In this case, too, naturally, the totals of the mass numbers must be equal on both sides of the arrow, and the charge numbers must balance likewise.

However, instead of producing the results thus indicated, the nuclear fission of this uranium atom may produce from the latter, for instance, the strontium atom $_{38}Sr^{88}$ and the xenon atom $_{54}Xe^{146}$, plus two neutrons, or atoms of two different elements and a different number of neutrons. In fact, in such fissions the formation of a very considerable number of different elements has been observed. The precise way in which the uranium nucleus splits (in which fission occurs) is a matter of chance to a certain extent.

We have thus obtained a general bird’s eye view of the different possibilities of nuclear transmutation. Neutrons can be used to transmute all nuclei; charged particles are most suitable for producing a transmutation of the lighter nuclear species. Among the charged particles which primarily enter into consideration are protons and deuterons, which are capable of being sufficiently accelerated in electric fields, as well as the alpha particles, both natural and artificially produced by an intensive acceleration of helium nuclei in electric fields. Finally, there are several nuclei which can be heated to such a high degree by a sufficiently energy-rich gamma-ray photon that they emit a neutron; this emission changes their mass only, but not the chemical properties of that particular atom.